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Module 05

Tunning Machine

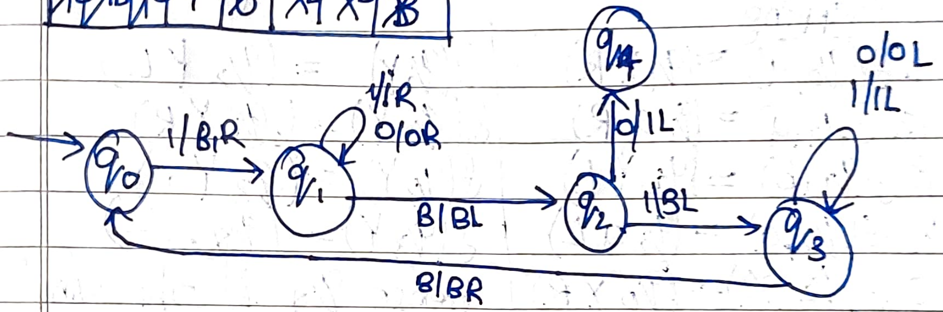
Unary (addition)

01. $f(m,n)$ { $m-n$ if $m>n$; 0 if $m<n$ }

1	1	1	1	0	1	1	B

Replace 1 By B and move right till you get back and move right till you get back, Dont Change blank Symbol and move left Replace 1 by Blank and move left until you get blank, process Repeats till you get Ans.

~~1~~B	~~1~~B	~~1~~B	1	~~0~~	~~X~~B	~~X~~B	B



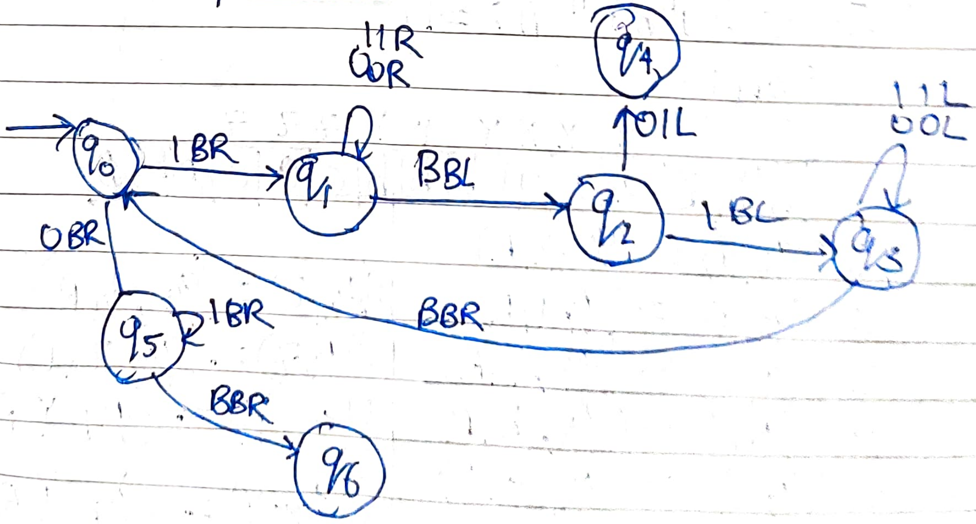
02-

B ~~0~~	B ~~0~~	B ~~1~~	B ~~0~~	B ~~0~~	B ~~0~~	B ~~0~~	B

$m=4$
$n=2$

or you can use like

B	1	1	0	1	1	1	1	B



03- Check if $L = \{(01)^n / n >= 0\}$

Mapping: $\delta(q_0, 0) = (q_1, X, R)$
$\delta(q_1, 1) = (q_0, Y, R)$
$\delta(q_0, b) = (q_2, -, -)$ $F = \{q_2\}$

			X	Y	X	Y	X	Y			
Y	Y	Y	0	1	0	1	0	1	Y	Y	Y
			q_0	q_1	q_0	q_1	q_0	q_1	q_0		

04- $L = \{a^n b^n / n >= 1\}$ As an accepting device

Mapping

1. $\delta(q_0, a) = (q_1, X, R)$
2. $\delta(q_1, a) = (q_1, a, R)$
3. $\delta(q_1, b) = (q_2, Y, L)$
4. $\delta(q_2, a) = (q_2, a, L)$
5. $\delta(q_2, X) = (q_0, X, R)$
6. $\delta(q_1, Y) = (q_1, Y, R)$

$\delta(q_2, Y) = (q_2, Y, L)$
$\delta(q_0, Y) = (q_3, Y, R)$
$\delta(q_3, Y) = (q_3, Y, R)$
$\delta(q_3, Y) = (q_4, -, -)$
Hence q_4 is the final state

		X	X	X	Y	Y	Y			
b	b	a	a	a	b	b	b	B	B	B

Input

05- $L = \{a^n b^n c^n / n >= 1\}$

Mapping

			X	X	X	Y	Y	Y	Z	Z	Z			
b	b	b	a	a	a	b	b	b	c	c	c	b	b	b

1. $\delta(q_0, a) = (q_1, X, R)$
2. $\delta(q_1, a) = (q_1, a, R)$
3. $\delta(q_1, b) = (q_2, Y, R)$
4. $\delta(q_2, b) = (q_2, b, R)$
5. $\delta(q_2, c) = (q_3, Z, L)$
6. $\delta(q_3, b) = (q_3, a, L)$
7. $\delta(q_3, a) = (q_3, a, L)$
8. $\delta(q_3, Y) = (q_3, b, L)$
9. $\delta(q_3, X) = (q_0, X, R)$
10. $\delta(q_0, Y) = (q_1, Y, R)$

$\delta(q_2, Z) = (q_2, Z, R)$
$\delta(q_3, Z) = (q_3, Z, L)$
$\delta(q_0, Y) = (q_4, Y, R)$
$\delta(q_4, Y) = (q_4, Y, R)$
$\delta(q_4, Z) = (q_5, Z, R)$
$\delta(q_5, Z) = (q_5, Z, R)$
$\delta(q_5, b) = (q_0, -, -)$

q_b is the final State
$F = \{q_b\}$

Q6- $L = \{wcw^R \mid w \in \{a,b\}^*\}$

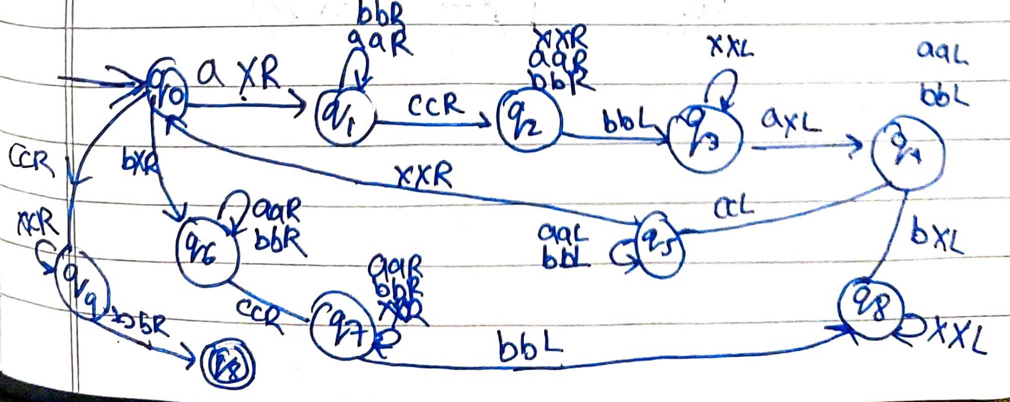
		X	X	X	X		X	X	X	X		
b̸	b̸	a	a	b	b	c	b	b	a	a	b̸	b̸

$\delta(q_0,a) = (q_1,a) = (q_1\ X\ R)$
$\delta(q_1\ a) = (q_1\ a\ R)$
$\delta(q_1\ b) = (q_1\ b\ R)$
$\delta(q_1\ c) = (q_2\ c\ R)$
$\delta(q_2,a) = (q_2\ a\ R)$
$\delta(q_2,\ b) = (q_2\ b\ R)$
$\delta(q_2\ b) = (q_3,\ b,\ L)$
$\delta(q_3\ X) = (q_3,\ X,\ L)$
$\delta(q_3\ a) = (q_4\ X\ L)$
$\delta(q_4,a) = (q_4\ a\ L)$
$\delta(q_4,b) = (q_4\ b\ R)$
$\delta(q_4,c) = (q_5\ c,\ L)$

$\delta(q_3,a) = (q_5\ a\ L)$
$\delta(q_5\ b) = (q_5\ b\ L)$
$\delta(q_5\ X) = (q_0\ X\ R)$
$\delta(q_0\ b) = (q_5\ X\ R)$
$\delta(q_6\ a) = (q_6\ a\ R)$
$\delta(q_6\ b) = (q_6\ b\ R)$
$\delta(q_6\ c) = (q_7\ c\ R)$
$\delta(q_7\ a) = (q_7\ a\ R)$
$\delta(q_7\ b) = (q_7\ b\ R)$
$\delta(q_7\ X) = (q_7\ X\ R)$
$\delta(q_7\ b) = (q_8\ b\ L)$
$\delta(q_8\ X) = (q_8\ X\ L)$
$\delta(q_8\ b) = (q_8\ X\ L)$
$\delta(q_2\ X) = (q_2\ X\ R)$

$\delta(q_0\ c) = (q_a\ c\ R)$
$\delta(q_a\ X) = (q_a\ X\ R)$
$\delta(q_a\ \cancel{b})\ (q_f\ -\ -)$

$F = \{q_f\}$



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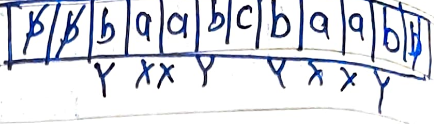
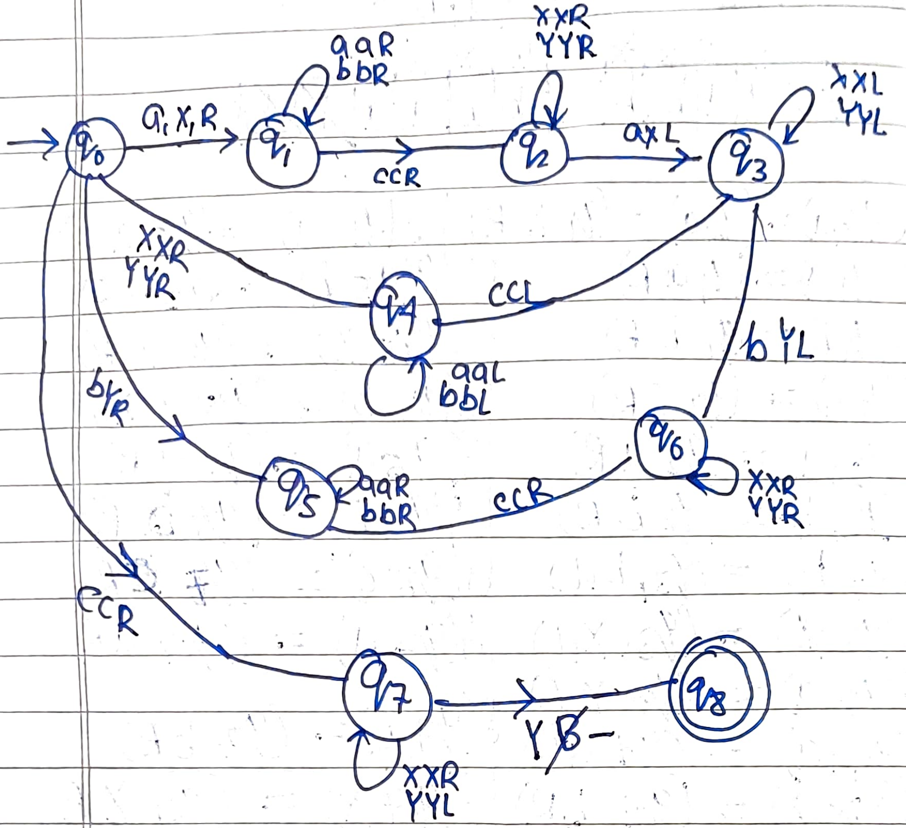
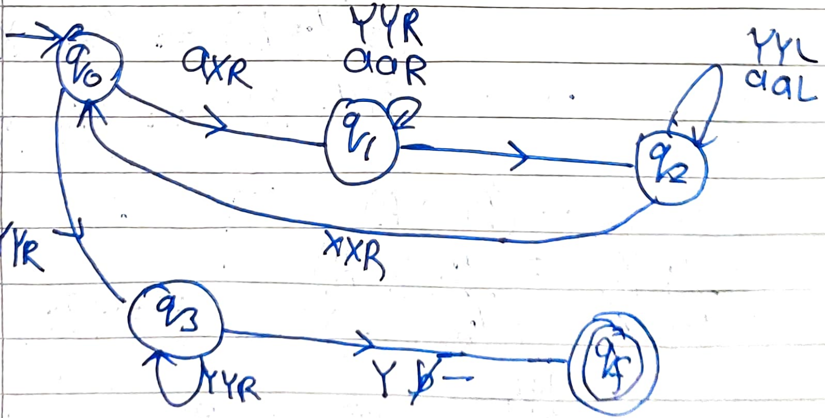
07 L= {wcw | w∈ {a,b}*}

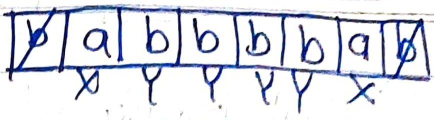


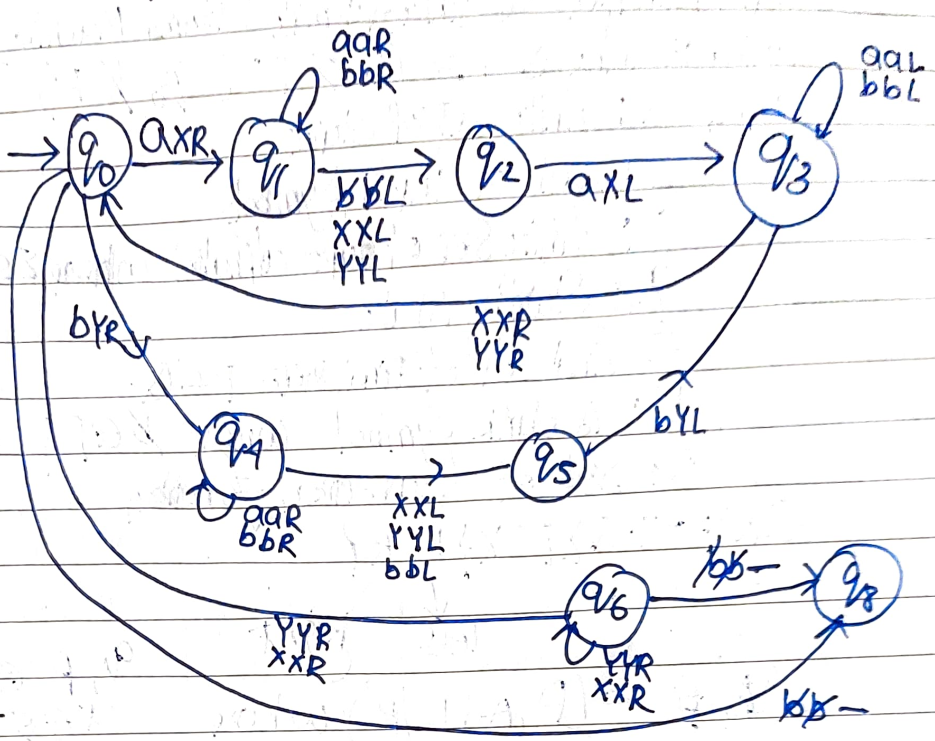


08- L={$a^n b^n$ | n>=1} - State diagram



Q9- $\{ww^R \mid w \in \{a,b\}^*\}$





Q10-

$L(M) = \{w \mid w \in \Sigma^*, q_0 w \vdash^* \alpha_1 q_f \alpha_2\}$

Where $\alpha_1 \alpha_2 \in F^*$, $q_f \in F$

formal definatio n of turning Machine

This is not Numarical & 4s Theory Part Just for Reference (Q10)

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Theory – Module 05

→ formal defination of finite Automate (Turing Machine)

Turing Machine is $M = (Q\ \Sigma\ \Gamma\ \delta\ q_0\ b\ F)$

where Q: finite set of states

Σ: set of Input Symbols

Γ: finite set of tape alphabet where $\Sigma \subseteq \Gamma$

q_0: Starting state

F: final set of Final state $F \subseteq Q$

b: is blank Symbol where $b \in \Gamma$

δ: Set of transition function

$Q \times \Sigma \rightarrow Q \times \Gamma \times \{L, R\}$

left Right

$\delta(q, a) = (P, A, L/R)$ where $q, P \in Q$, $a \in \Sigma$, $A \in \Gamma$

→ Instant String:

String of the form $\alpha_1 q \alpha_2$ where $\alpha_1 \alpha_2 \in \Gamma^*$, $q \in Q$.

Initial ID – $q_0 w$

If the current ID is – $\underbrace{x_1 x_2 \cdots x_{i-1}}_{\alpha_1} q \underbrace{x_i \cdots x_m}_{\alpha_2}$

$\alpha_1, \alpha_2 \in \Gamma^*$ $q \in Q$

Now if we have a move/mapping

$\delta(q, x_i) = (P A R)$, then

$x_1 x_2 \cdots x_{i-1}\, q\, x_i \cdots x_m \vdash x_1 x_2 \cdots x_{i-1} A P x_{i+1}$

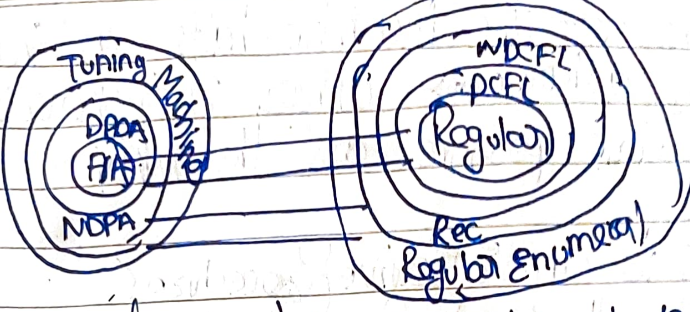
If we have Move mapping

$\delta(q, x_i) = (P A L)$ then

$x_1 x_2 \cdots x_{i-1}\, q\, x_i \cdots x_m \vdash x_1 x_2 \cdots x_{i-2} P$

Final id: $\alpha_1 q_f \alpha_2$ where $\alpha_1 \alpha_2 \in \Gamma^*$, $q_f \in F$

Formal defination of turing Machine | $L(M) = \{w \mid w \in \Sigma^*, q_0 w \vdash^* \alpha_1 q_f \alpha_2\}$ where $\alpha_1 \alpha_2 \in \Gamma^*$, $q_f \in F$



- Turning developed turing Machine in 1936.
- Church-Turning hypotesis - what Could naturally be called effective procedure can be regalized by a turning Machine
- Main Components of Turning Machine
(.) It has an infinite tape, the tape Can be infinite from both directions or one direction
- Apart from the same position of the tape, rest of the postions is blank. The tape is infinite but the non blank positions is only finite

We Can look turning Machine As an Input/output device or a accepting device, where When we look it as an accepting device then it accepts an Type 0 languages

when We look it as an I/O device the at first the input is Present in the non-blank portion of tape

b̸	b̸	b̸	a	a	a	b̸	b̸	b̸

The Machine is in an State q, It Reads the Input symbol a Next it Can be rewritten and the Current State Can change head and Can Move left/Right

There are Some Symbols Called Tape Symbols

Mapping: $\delta(q, a) = P(P, A, b̸, R)$

Next State

A Reservive as a

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At the last the Machine halts, at that Moment wherever where ever there is non-blank postions are the output

Q- what is an Effective Procedure?
Usually the word Procedure and algorithm, They are used in the Same Manner, but There is difference Between them.

Procedure- Essencially its an Programm, it tells us what is the Next Step needs to be done, More Importantly a procedure can have halt or no halts

Algorithm- It is also an Programm it always halts and gives the answer

Example
first problem
Input is an numbear n, we needs to check the input does have certain properties or not

-we can Easily write an algorithm for that

2nd Problem Now the Input is an numbear n, we need to check after the numbear n is there a numbear is there any number which have certain fo

-Here if the perticular numbers Exist and we know that in that Case definetly an algorithm is possible

It will check n+1, if it get the numbers with the

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required Properties then the Machine halts and gives the result yes. if no then the it will check for n+2, n+3 and so on. So the Ammount of time the Machine halts and gives the answer yes as we know the number is Existing.

On the other hand, Suppose the Required numbers does not Exist, then the Machine will never halt and it keeps on Searching, and Result in Infinite loop

Algorithm — When a turning Machine halts and gives results for some program then it is called algorithm [Informal defination]

Parity Checking - Construction of TM as i/p - o/p devices

b̸	b̸	b̸	1	0	1	1	0	0	1	1	b̸	b̸	b̸

for any binary String if the number of 1's is even then TM prints Even, otherwise odd.

Mapping $\delta(q_0, 0) = (q_0\ X R)$
$\delta(q_0, 1) = (q_1\ X R)$
$\delta(q_1, 0) = (q_1\ X R)$
$\delta(q_1, 1) = (q_0\ X R)$
$\delta(q_0\ b̸) = (\text{Halt Even} -)$
$\delta(q_1\ b̸) = (\text{Halt odd} -)$

— defenitaly it could be done by FSA

Notes This is only for Understanding how Turing Machine works.

→ A Paranthesis Checker
Here we have to check well formed strings

b̸	b̸	[	A	C	C	)	C	)	)	C	)	AS	b̸	b̸	]

Mapping -

$\delta(q_0, C) = (q_0\ C\ R)$
$\delta(q_0\ D) = (q_1\ X\ L)$
$\delta(q_1\ C) = (q_0\ X\ R)$
$\delta(q_0\ X) = (q_0\ X\ R)$
$\delta(q_1, X) = (q_1\ X\ L)$
$\delta(q_0\ A) = (q_2\ A\ L)$
$\delta(q_2\ X) = (q_2\ X\ L)$
$\delta(q_2, A) = (\text{Halt}, \text{Yes}\ -)$

$\delta(q_1\ A) = (\text{Halt}, \text{No}\ -)$

$\delta(q_2\ C) = (\text{Halt No}\ -)$

→ Uniary to Binary

Uniary Representation of 5 - 11111
7 - 1111111

b	b	b	b	1	1	1	1	1	b	b	b	b

Y = 1
Z = 0

Mapping

$\delta(q_0\ 1) = (q_1\ X\ R)$
$\delta(q_1\ 1) = (q_0\ 1\ R)$
$\delta(q_1\ b) = (q_3\ b\ L)$
$\delta(q_3\ 1) = (q_3\ 1\ L)$
$\delta(q_3\ X) = (q_3\ X\ L)$
$\delta(q_3\ b) = (q_4\ Y\ R)$
$\delta(q_4\ 1) = (q_1\ X\ R)$
$\delta(q_4\ X) = (q_4\ X\ R)$
$\delta(q_1\ X) = (q_1\ X\ R)$
$\delta(q_0\ X) = (q_0\ X\ R)$
$\delta(q_0\ b) = (q_2\ b\ L)$
$\delta(q_2\ 1) = (q_2\ 1\ L)$
$\delta(q_2\ X) = (q_2\ X\ L)$
$\delta(q_2\ X) = (q_2\ X\ L)$

$\delta(q_2\ b) = (q_4\ Z\ R)$
$\delta(q_4\ Y) = (q_4\ Y\ R)$
$\delta(q_3\ 1) = (q_3\ 1\ L)$
$\delta(q_3\ Y) = (q_3\ Y\ L)$
$\delta(q_3\ Z) = (q_3\ Z\ L)$
$\delta(q_2\ Z) = (q_2\ Z\ L)$
$\delta(q_4\ b) = (\text{Halt}\ b)$

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→ 1's Compliment

Mapping

$\delta(q_0, 0) = (q_1\ 1\ R)$
$\delta(q_0\ 1) = (q_1\ 0\ R)$
$\delta(q_1\ 0) = (q_0\ 1\ R)$
$\delta(q_1\ 1) = (q_0\ 0\ R)$
$\delta(q_0\ \cancel{b}) = (\text{Halt}\ \cancel{b}\ -)$
$\delta(q_1\ \cancel{b}) = (\text{Halt}\ \cancel{b}\ -)$

→ 2's Compliment

Mapping

$\delta(q_0\ 0) = (q_0\ 0\ R)$
$\delta(q_0\ 1) = (q_0\ 1\ R)$
$\delta(q_0\ \cancel{b}) = (q_1\ \cancel{b}\ L)$
$\delta(q_1\ 0) = (q_1\ 0\ L)$
$\delta(q_1\ 1) = (q_2\ 1\ L)$
$\delta(q_2\ 0) = (q_2\ 1\ L)$
$\delta(q_2\ 1) = (q_2\ 0, L)$
$\delta(q_2\ \cancel{b}) = (\text{Halt}, \cancel{b}\ -)$

$\cancel{b}$ 0 1 1 0 1

$\cancel{b}$	$\cancel{b}$	1	0	0	1	0	1	$\cancel{b}$	$\cancel{b}$

$q_0\ q_0\ q_0\ q_0\ q_0\ q_0\ q_0\ q_0$

$q_2\ q_2\ q_2\ q_2\ q_2\ q_2\ q_1$

$\cancel{b}$	$\cancel{b}$	0	1	1	0	1	0	$\cancel{b}$	$\cancel{b}$

$q_0\ q_0\ q_0\ q_0\ q_0\ q_0\ q_0\ q_0$ ←

$q_2\ q_2\ q_2\ q_2\ q_2\ q_2\ q_1$
$\cancel{b}$ 1 0 0 1 1 0

100101
⇓
011010
+1
011011

011010
⇓
100101
+1
100110